Proof of Conjecture 3: Strict Monotonicity of ρ_p for Hadamard Matrices

Problem Statement

When studying the stability of nonlinear PDE of the form $\partial_t u + A(u)\partial_x u + B(u) = 0$ on $[0, L]$, the following stabilizing quantities arise:

$$\rho_p(M) := \inf\{\|\Delta M \Delta^{-1}\|_p \mid \Delta = \mathrm{diag}(\delta_1, \ldots, \delta_n),\ \delta_i > 0\},$$

where $\|\cdot\|_p$ is the induced ℓ^p-operator norm.

Conjecture 1 (Conjecture 3)**.** *For any $n > 6$, and any $p > 2$ ($n, p \in \mathbb{N}$), there exists $M \in \mathbb{R}^{n\times n}$ such that*

$$\rho_p(M) < \rho_{p+1}(M).$$

Overview of the Proof

Our strategy is to use **Sylvester–Hadamard matrices**. For an $m\times m$ Hadamard matrix H satisfying $H^\top H = mI$, we establish the exact formula $\rho_r(H) = m^{1-1/r}$ for all $r \geq 2$ by combining:

- An **upper bound** via the Riesz–Thorin interpolation theorem (interpolating between the ℓ^2 and ℓ^∞ operator norms);
- A matching **lower bound** via a test-vector argument using the rows of H.

Since $m^{1-1/r}$ is strictly increasing in r (for $m \geq 2$), we get $\rho_p(H) < \rho_{p+1}(H)$ for all $p \geq 2$. Embedding H into larger dimensions via zero-padding completes the proof for all $n > 6$.

1 The Hadamard Matrix and Its Operator Norms

For clarity, write H_m for the Sylvester–Hadamard matrix of order $m = 2^k$. Thus $H_1 = (1)$, and for $m = 2n$,

$$H_m = H_{2n} = \begin{pmatrix} H_n & H_n \\ H_n & -H_n \end{pmatrix}.$$

We prove the following properties.

(a) H is symmetric and has only ± 1 entries

We prove by induction on k that H_{2^k} is symmetric and every entry of H_{2^k} is equal to $+1$ or -1.

For $k = 0$, we have $H_1 = (1)$, which is symmetric and has only $+1$ entries.

Assume now that H_n is symmetric and has only ± 1 entries. Consider

$$H_{2n} = \begin{pmatrix} H_n & H_n \\ H_n & -H_n \end{pmatrix}.$$

Every block H_n and $-H_n$ has only ± 1 entries, so H_{2n} also has only ± 1 entries. Moreover, since $H_n^\top = H_n$, we have

$$H_{2n}^\top = \begin{pmatrix} H_n^\top & H_n^\top \\ H_n^\top & (-H_n)^\top \end{pmatrix} = \begin{pmatrix} H_n & H_n \\ H_n & -H_n \end{pmatrix} = H_{2n}.$$

Thus H_{2n} is symmetric. By induction, H_m is symmetric and has only ± 1 entries for every $m = 2^k$.

(b) $H^\top H = H^2 = mI_m$

Since H_m is symmetric by part (a), it suffices to prove $H_m^2 = mI_m$.

We prove this by induction on $m = 2^k$. For $m = 1$, $H_1^2 = (1)^2 = (1) = I_1$, so the claim holds.

Assume now that for some $n = 2^{k-1}$, we have $H_n^2 = nI_n$. Let $m = 2n$. Then

$$H_{2n}^2 = \begin{pmatrix} H_n & H_n \\ H_n & -H_n \end{pmatrix} \begin{pmatrix} H_n & H_n \\ H_n & -H_n \end{pmatrix} = \begin{pmatrix} 2H_n^2 & 0 \\ 0 & 2H_n^2 \end{pmatrix}.$$

Using the induction hypothesis $H_n^2 = nI_n$, we get

$$H_{2n}^2 = \begin{pmatrix} 2nI_n & 0 \\ 0 & 2nI_n \end{pmatrix} = 2nI_{2n}.$$

Since $m = 2n$, this is exactly $H_m^2 = mI_m$. Thus by induction $H^2 = mI_m$, and since $H^\top = H$, we have $H^\top H = H^2 = mI_m$.

(c) Operator norms of H

Spectral norm. Since $H^\top H = mI_m$, every eigenvalue of $H^\top H$ equals m. Hence

$$\|H\|_{2\to 2} = \sqrt{\lambda_{\max}(H^\top H)} = \sqrt{m}.$$

ℓ^1 operator norm. Since $|H_{ij}| = 1$ for every i, j, each column has absolute column sum $\sum_{i=1}^m |H_{ij}| = m$. Hence $\|H\|_{1\to 1} = m$.

ℓ^∞ operator norm. Similarly, each row has absolute row sum $\sum_{j=1}^m |H_{ij}| = m$. Hence $\|H\|_{\infty\to\infty} = m$.

(d) $Hh_i = me_i$

Let h_i be the ith row of H, viewed as a column vector. Since H is symmetric, $h_i = H^\top e_i = He_i$. Using part (b),

$$Hh_i = H(He_i) = H^2 e_i = mI_m e_i = me_i.$$

2 Upper Bound on $\|H\|_{r\to r}$ via Riesz–Thorin Interpolation

We use the following standard result.

Theorem 1 (Riesz–Thorin interpolation theorem)**.** *Let (Ω, μ) be a σ-finite measure space, and let $1 \le p_0, p_1, q_0, q_1 \le \infty$. For $0 < \theta < 1$, define p_θ, q_θ by*

$$\frac{1}{p_\theta} = \frac{1-\theta}{p_0} + \frac{\theta}{p_1}, \qquad \frac{1}{q_\theta} = \frac{1-\theta}{q_0} + \frac{\theta}{q_1},$$

with the convention $1/\infty = 0$. Suppose $T : L^{p_0}(\Omega) + L^{p_1}(\Omega) \to L^{q_0}(\Omega) + L^{q_1}(\Omega)$ is linear, and that for $i = 0, 1$,

$$T\big(L^{p_i}(\Omega)\big) \subseteq L^{q_i}(\Omega), \qquad \|Tf\|_{q_i} \le M_i \|f\|_{p_i} \quad \textit{for all } f \in L^{p_i}(\Omega).$$

Then $T\big(L^{p_\theta}(\Omega)\big) \subseteq L^{q_\theta}(\Omega)$ and $\|T\|_{L^{p_\theta}\to L^{q_\theta}} \le M_0^{1-\theta} M_1^{\theta}$.

Let $\Omega = \{1, \dots, m\}$ with counting measure μ. Then $\mu(\Omega) = m < \infty$, so (Ω, μ) is σ-finite, and $L^p(\Omega, \mu) = \ell_p^m$ for every $1 \le p \le \infty$.

Define $T : \mathbb{C}^m \to \mathbb{C}^m$ by $Tx = Hx$. Since $L^2(\Omega) + L^\infty(\Omega) = \mathbb{C}^m$ (all L^p spaces coincide on a finite set), T is one linear map on the common domain, and its endpoint actions on L^2 and L^∞ are restrictions of the same operator, verifying the compatibility hypothesis.

From Section 1, $\|H\|_{2\to 2} = \sqrt{m}$ and $\|H\|_{\infty\to\infty} = m$. We apply Riesz–Thorin with

$$p_0 = q_0 = 2, \quad p_1 = q_1 = \infty, \quad M_0 = \sqrt{m}, \quad M_1 = m.$$

For $r = 2$, $\|H\|_{2\to 2} = \sqrt{m} = m^{1-1/2}$, and the desired estimate holds.

For $r > 2$, choose $\theta = 1 - 2/r$. Since $r > 2$, we have $0 < \theta < 1$. Then

$$\frac{1}{p_\theta} = \frac{1-\theta}{2} + \frac{\theta}{\infty} = \frac{1-\theta}{2} = \frac{1}{r},$$

and similarly $1/q_\theta = 1/r$. So $p_\theta = q_\theta = r$. Applying Riesz–Thorin:

$$\|H\|_{r\to r} \le (\sqrt{m})^{1-\theta} m^{\theta} = (\sqrt{m})^{2/r} m^{1-2/r} = m^{1/r} m^{1-2/r} = m^{1-1/r}.$$

Therefore, $\|H\|_{r\to r} \le m^{1-1/r}$ for every real $r \ge 2$.

3 Lower Bound: Diagonal Scaling Cannot Reduce $\rho_r(H)$ Below $m^{1-1/r}$

Proposition 2. *For every real $r \geq 2$ and every positive diagonal matrix $D = \operatorname{diag}(d_1, \ldots, d_m)$ with $d_i > 0$:*

$$\|DHD^{-1}\|_{r\to r} \geq m^{1-1/r},$$

and hence $\rho_r(H) \geq m^{1-1/r}$.

Proof. Fix a positive diagonal matrix $D = \operatorname{diag}(d_1, \ldots, d_m)$ with $d_i > 0$. Let $A := DHD^{-1}$.

Choose an index $i^* \in \{1, \ldots, m\}$ such that $d_{i^*} = \max_{1\leq j\leq m} d_j$. Let $h_{i^*} \in \{\pm 1\}^m$ denote the i^*-th row of H, viewed as a column vector. By Section 1(d), $Hh_{i^*} = me_{i^*}$.

Define the test vector $y := Dh_{i^*}$. Since D is invertible and $h_{i^*} \neq 0$, we have $y \neq 0$.

We compute

$$Ay = DHD^{-1}Dh_{i^*} = DHh_{i^*} = D(me_{i^*}) = md_{i^*}e_{i^*}.$$

Therefore $\|Ay\|_r = md_{i^*}$.

Since every entry of h_{i^*} has absolute value 1,

$$\|y\|_r = \|Dh_{i^*}\|_r = \left(\sum_{j=1}^{m} d_j^r\right)^{1/r}.$$

Because $d_j \leq d_{i^*}$ for every j, we have $\sum_{j=1}^m d_j^r \leq md_{i^*}^r$, so $\|y\|_r \leq m^{1/r}d_{i^*}$.

By the definition of the induced operator norm,

$$\|A\|_{r\to r} \geq \frac{\|Ay\|_r}{\|y\|_r} = \frac{md_{i^*}}{\left(\sum_{j=1}^m d_j^r\right)^{1/r}} \geq \frac{md_{i^*}}{m^{1/r}d_{i^*}} = m^{1-1/r}.$$

Since the positive diagonal matrix D was arbitrary, taking the infimum over all such D gives

$$\rho_r(H) = \inf_{\substack{D=\operatorname{diag}(d_1,\ldots,d_m)\\ d_i>0}} \|DHD^{-1}\|_{r\to r} \geq m^{1-1/r}. \qquad \square$$

4 Exact Formula and Strict Monotonicity of $\rho_r(H)$

Theorem 3. *Let H be an $m \times m$ Sylvester–Hadamard matrix with $m = 2^k$ and $k \geq 1$. Then for every real $r \geq 2$:*

$$\rho_r(H) = m^{1-1/r}.$$

Moreover, the function $r \mapsto \rho_r(H)$ is strictly increasing on $[2, \infty)$.

Proof. **Part (a).** Fix any real number $r \geq 2$. By the upper bound from Section 2, using $D = I$:

$$\rho_r(H) \leq \|H\|_{r\to r} \leq m^{1-1/r}.$$

By the lower bound from Section 3:

$$\rho_r(H) \geq m^{1-1/r}.$$

Therefore $\rho_r(H) = m^{1-1/r}$.

Part (b). Since $k \geq 1$, we have $m \geq 2$, so $m > 1$.

Let $2 \leq r < s$. Then $1/r > 1/s$, so $1 - 1/r < 1 - 1/s$. Since $m > 1$, the exponential function $t \mapsto m^t$ is strictly increasing. Hence

$$\rho_r(H) = m^{1-1/r} < m^{1-1/s} = \rho_s(H).$$

In particular, for every real $p \geq 2$, taking $r = p$ and $s = p + 1$:

$$\rho_p(H) = m^{1-1/p} < m^{1-1/(p+1)} = \rho_{p+1}(H),$$

where the strict inequality follows because

$$\left(1 - \frac{1}{p+1}\right) - \left(1 - \frac{1}{p}\right) = \frac{1}{p} - \frac{1}{p+1} = \frac{1}{p(p+1)} > 0. \qquad \square$$

5 Embedding into Larger Dimensions and Proof of Conjecture 3

Part (a): Block-diagonal embedding preserves ρ_r

Lemma 4. *For every $A \in \mathbb{R}^{m\times m}$, every integer $s \geq 0$, and every real $r \geq 1$:*

$$\|\operatorname{diag}(A, 0_s)\|_{r\to r} = \|A\|_{r\to r}.$$

Proof. Write $x = (x_1, x_2) \in \mathbb{R}^m \oplus \mathbb{R}^s$. Then $\|x\|_r^r = \|x_1\|_r^r + \|x_2\|_r^r$, so $\|x_1\|_r \leq \|x\|_r$.

Upper bound: $\|\operatorname{diag}(A, 0_s)x\|_r = \|(Ax_1, 0)\|_r = \|Ax_1\|_r \leq \|A\|_{r\to r}\|x_1\|_r \leq \|A\|_{r\to r}\|x\|_r$. Taking the supremum over $x \neq 0$ gives $\|\operatorname{diag}(A, 0_s)\|_{r\to r} \leq \|A\|_{r\to r}$.

Lower bound: Since the ℓ^r-unit sphere in $\mathbb{R}^m$ is compact and $u \mapsto \|Au\|_r$ is continuous, the supremum $\|A\|_{r\to r} = \sup_{\|u\|_r=1} \|Au\|_r$ is attained at some $u^* \in \mathbb{R}^m$. Then

$$\|\operatorname{diag}(A, 0_s)\|_{r\to r} \geq \frac{\|(Au^*, 0)\|_r}{\|(u^*, 0)\|_r} = \frac{\|Au^*\|_r}{\|u^*\|_r} = \|A\|_{r\to r}. \qquad \square$$

Proposition 5. *For every integer $n \geq m$ and every real $r \geq 1$: $\rho_r(M_n) = \rho_r(H)$, where $M_n = \operatorname{diag}(H, 0_{n-m})$.*

Proof. Any positive diagonal $D \in \mathbb{R}^{n\times n}$ can be written as $D = \operatorname{diag}(D_1, D_2)$ with D_1 and D_2 positive diagonal. Then $DM_nD^{-1} = \operatorname{diag}(D_1HD_1^{-1}, 0_{n-m})$. By Lemma **??**, $\|DM_nD^{-1}\|_{r\to r} = \|D_1HD_1^{-1}\|_{r\to r}$.

As D ranges over all positive diagonal $n \times n$ matrices, D_1 ranges over all positive diagonal $m \times m$ matrices (independently of D_2). Hence

$$\rho_r(M_n) = \inf_D \|DM_nD^{-1}\|_{r\to r} = \inf_{D_1} \|D_1HD_1^{-1}\|_{r\to r} = \rho_r(H). \qquad \square$$

Part (b): Proof of Conjecture 3

We use the explicit Hadamard matrices:

$$H_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}.$$

A direct computation gives $H_4H_4^\top = 4I_4$ (each row has squared norm 4, and distinct rows have inner product 0). We also define H_8 via the Sylvester construction:

$$H_8 = \begin{pmatrix} H_4 & H_4 \\ H_4 & -H_4 \end{pmatrix},$$

which satisfies $H_8H_8^\top = 8I_8$ by the block multiplication $H_8H_8^\top = \operatorname{diag}(2H_4H_4^\top, 2H_4H_4^\top) = 8I_8$.

From Section 1, for any $m \times m$ Hadamard matrix H (with $HH^\top = mI$), we have $H^\top H = mI$ as well, since $H(H^\top/m) = I$ implies H is invertible with $H^{-1} = H^\top/m$, hence $H^\top H = mI$.

By Theorem **??**, $\rho_q(H) = m^{1-1/q}$ for every $q \geq 2$.

Let $n > 6$ and $p > 2$ be integers ($p \geq 3$).

Case 1: $n = 7$**.** Take $M = \operatorname{diag}(H_4, 0_3) \in \mathbb{R}^{7\times 7}$. By Part (a), $\rho_p(M) = \rho_p(H_4) = 4^{1-1/p}$ and $\rho_{p+1}(M) = 4^{1-1/(p+1)}$. Since $1 - 1/p < 1 - 1/(p+1)$ and $4 > 1$:

$$\rho_p(M) = 4^{1-1/p} < 4^{1-1/(p+1)} = \rho_{p+1}(M).$$

Case 2: $n \geq 8$**.** Take $M = \operatorname{diag}(H_8, 0_{n-8}) \in \mathbb{R}^{n\times n}$ (with the zero block omitted when $n = 8$). By Part (a), $\rho_p(M) = 8^{1-1/p}$ and $\rho_{p+1}(M) = 8^{1-1/(p+1)}$. Since $8 > 1$:

$$\rho_p(M) = 8^{1-1/p} < 8^{1-1/(p+1)} = \rho_{p+1}(M).$$

In both cases, for every integer $n > 6$ and every integer $p > 2$, we have exhibited $M \in \mathbb{R}^{n\times n}$ with $\rho_p(M) < \rho_{p+1}(M)$. This proves Conjecture 3. $\square$

References

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